

Supplementary Materials

Impact of Scanner Vendor Harmonization on Spinal Cord Quantitative MRI Biomarkers: A Systematic Benchmark of Five Statistical Correction Strategies

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Supplementary Table S1

Per-biomarker evaluation table: all metrics for 8 biomarkers x 5 methods x 2 cohorts (96 rows, 11 columns). Metrics include vendor F-statistic, vendor p-value, vendor eta-squared, vendor ICC, age semi-partial R-squared, sex semi-partial R-squared, and pre-post Pearson r.

Abbreviations: ComBat = Combatting batch effects; CovBat = Covariance Batch harmonization; LME/FE = linear mixed-effects / fixed-effects hybrid model; RELIEF = Removal of Latent Inter-scanner Effects via independent components; ICC = intraclass correlation coefficient; eta2 = eta-squared; R2 = semi-partial R-squared.

Cohort	Method	Biomarker	N	Vendor F	Vendor p	Vendor eta2	Vendor ICC	Age R2	Sex R2	r(pre,post)
HC	Original	T2w_CSA	203.0	5.610	0.0043	0.0531	0.0638	7.95e-04	0.1047	1.000
HC	Original	GM_CSA_mm2	203.0	21.61	3.18e-09	0.1777	0.2335	0.0018	0.0894	1.000
HC	Original	MTR	194.0	18.44	4.76e-08	0.1618	0.2124	0.0117	3.00e-04	1.000
HC	Original	MTsat	190.0	47.71	1.81e-17	0.3379	0.4245	0.0057	0.0023	1.000
HC	Original	FA	198.0	36.08	4.67e-14	0.2701	0.3470	2.54e-04	0.0044	1.000
HC	Original	MD	198.0	59.02	9.09e-21	0.3771	0.4678	1.13e-04	6.83e-04	1.000
HC	Original	AD	198.0	160.6	5.98e-42	0.6223	0.7075	2.97e-04	7.37e-06	1.000
HC	Original	RD	198.0	30.06	4.19e-12	0.2356	0.3057	2.40e-05	0.0015	1.000
HC	ComBat	T2w_CSA	203.0	0.0479	0.9532	4.79e-04	-0.0143	9.12e-07	0.1007	0.9721
HC	ComBat	GM_CSA_mm2	203.0	0.0407	0.9601	4.07e-04	-0.0144	2.61e-06	0.0910	0.9129
HC	ComBat	MTR	194.0	0.0354	0.9652	3.71e-04	-0.0151	0.0074	6.45e-04	0.8510
HC	ComBat	MTsat	190.0	0.0038	0.9962	4.07e-05	-0.0160	2.73e-05	0.0070	0.7880
HC	ComBat	FA	198.0	0.0153	0.9849	1.56e-04	-0.0151	7.76e-04	0.0131	0.8320
HC	ComBat	MD	198.0	0.0488	0.9523	5.01e-04	-0.0146	0.0111	0.0047	0.7615
HC	ComBat	AD	198.0	0.0671	0.9351	6.88e-04	-0.0143	0.0208	0.0027	0.5889
HC	ComBat	RD	198.0	0.0351	0.9655	3.60e-04	-0.0148	0.0074	0.0056	0.8270
HC	ComBat-joint	T2w_CSA	188.0	0.0563	0.9453	6.08e-04	-0.0153	3.97e-05	0.1218	0.9747
HC	ComBat-joint	GM_CSA_mm2	188.0	0.1041	0.9012	0.0011	-0.0145	1.98e-05	0.1013	0.9200
HC	ComBat-joint	MTR	188.0	0.0634	0.9386	6.85e-04	-0.0152	1.82e-04	8.87e-04	0.8162
HC	ComBat-joint	MTsat	188.0	0.0496	0.9516	5.36e-04	-0.0154	3.29e-05	0.0056	0.8106
HC	ComBat-joint	FA	188.0	0.0336	0.9670	3.63e-04	-0.0157	8.79e-04	0.0136	0.8456
HC	ComBat-joint	MD	188.0	0.0125	0.9876	1.35e-04	-0.0160	0.0206	0.0088	0.7190
HC	ComBat-joint	AD	188.0	0.0262	0.9741	2.84e-04	-0.0158	0.0330	0.0054	0.5448
HC	ComBat-joint	RD	188.0	0.0180	0.9822	1.94e-04	-0.0159	0.0146	0.0101	0.8193
HC	CovBat	T2w_CSA	188.0	0.0345	0.9661	3.72e-04	-0.0156	5.00e-06	0.1279	0.9399
HC	CovBat	GM_CSA_mm2	188.0	0.0195	0.9807	2.11e-04	-0.0159	3.69e-06	0.1098	0.8585
HC	CovBat	MTR	188.0	0.0391	0.9617	4.22e-04	-0.0156	1.33e-06	0.0011	0.7713
HC	CovBat	MTsat	188.0	0.0144	0.9857	1.55e-04	-0.0160	1.20e-06	0.0055	0.7934
HC	CovBat	FA	188.0	0.0372	0.9635	4.02e-04	-0.0156	3.57e-04	0.0123	0.8377
HC	CovBat	MD	188.0	0.0087	0.9913	9.42e-05	-0.0161	0.0221	0.0089	0.7163
HC	CovBat	AD	188.0	0.0210	0.9792	2.27e-04	-0.0159	0.0393	0.0062	0.5251
HC	CovBat	RD	188.0	0.0135	0.9866	1.46e-04	-0.0160	0.0146	0.0097	0.8147
HC	RELIEF	T2w_CSA	188.0	0.1204	0.8866	0.0013	-0.0142	4.30e-05	0.1317	0.9429
HC	RELIEF	GM_CSA_mm2	188.0	0.0505	0.9508	5.45e-04	-0.0154	2.22e-05	0.1132	0.8663
HC	RELIEF	MTR	188.0	0.0574	0.9442	6.21e-04	-0.0153	1.84e-04	8.97e-04	0.8040

HC	RELIEF	MTsat	188.0	0.0291	0.9713	3.15e-04	-0.0157	3.39e-05	0.0058	0.7988
HC	RELIEF	FA	188.0	0.0206	0.9797	2.22e-04	-0.0159	8.88e-04	0.0137	0.8400
HC	RELIEF	MD	188.0	0.0072	0.9928	7.79e-05	-0.0161	0.0207	0.0089	0.7211
HC	RELIEF	AD	188.0	0.0328	0.9677	3.55e-04	-0.0157	0.0340	0.0056	0.5310
HC	RELIEF	RD	188.0	0.0049	0.9951	5.31e-05	-0.0161	0.0147	0.0102	0.8213
HC	LME	T2w_CSA	203.0	0.0471	0.9540	4.71e-04	-0.0143	1.72e-05	0.1024	0.9778
HC	LME	GM_CSA_mm2	203.0	0.0400	0.9608	4.00e-04	-0.0144	8.94e-07	0.0926	0.9148
HC	LME	MTR	194.0	0.0345	0.9661	3.61e-04	-0.0152	0.0158	4.98e-06	0.9147
HC	LME	MTsat	190.0	0.0037	0.9963	3.99e-05	-0.0160	8.23e-06	0.0096	0.8104
HC	LME	FA	198.0	0.0150	0.9851	1.54e-04	-0.0151	2.67e-04	0.0136	0.8498
HC	LME	MD	198.0	0.0194	0.9807	1.99e-04	-0.0151	0.0102	0.0066	0.7843
HC	LME	AD	198.0	0.0465	0.9546	4.77e-04	-0.0147	0.0196	0.0034	0.6038
HC	LME	RD	198.0	0.0345	0.9661	3.54e-04	-0.0148	0.0060	0.0076	0.8650
ALL	Original	T2w_CSA	267.0	5.845	0.0033	0.0424	0.0516	4.19e-04	0.0857	1.000
ALL	Original	GM_CSA_mm2	267.0	27.06	2.04e-11	0.1701	0.2265	0.0027	0.0641	1.000
ALL	Original	MTR	254.0	27.54	1.54e-11	0.1800	0.2387	0.0292	0.0012	1.000
ALL	Original	MTsat	249.0	69.34	1.31e-24	0.3605	0.4516	0.0128	0.0060	1.000
ALL	Original	FA	260.0	50.91	2.37e-19	0.2838	0.3654	0.0040	0.0071	1.000
ALL	Original	MD	260.0	93.61	2.88e-31	0.4215	0.5166	8.64e-04	0.0053	1.000
ALL	Original	AD	260.0	250.6	4.17e-61	0.6611	0.7423	1.88e-04	0.0028	1.000
ALL	Original	RD	260.0	45.73	1.02e-17	0.2625	0.3404	0.0032	0.0059	1.000
ALL	ComBat	T2w_CSA	267.0	0.0029	0.9972	2.16e-05	-0.0113	7.23e-05	0.0927	0.9733
ALL	ComBat	GM_CSA_mm2	267.0	0.0041	0.9959	3.12e-05	-0.0113	2.35e-04	0.0771	0.9110
ALL	ComBat	MTR	254.0	0.0749	0.9279	5.96e-04	-0.0110	0.0306	0.0033	0.8615
ALL	ComBat	MTsat	249.0	0.0055	0.9946	4.44e-05	-0.0121	0.0033	0.0064	0.7797
ALL	ComBat	FA	260.0	0.0218	0.9784	1.70e-04	-0.0114	0.0147	0.0105	0.8252
ALL	ComBat	MD	260.0	0.0319	0.9686	2.49e-04	-0.0113	0.0153	0.0091	0.7346
ALL	ComBat	AD	260.0	0.0215	0.9787	1.68e-04	-0.0114	0.0136	0.0084	0.5636
ALL	ComBat	RD	260.0	0.0320	0.9685	2.49e-04	-0.0113	0.0157	0.0084	0.8099
ALL	ComBat-joint	T2w_CSA	246.0	0.0022	0.9978	1.82e-05	-0.0123	3.96e-04	0.1141	0.9742
ALL	ComBat-joint	GM_CSA_mm2	246.0	0.0100	0.9901	8.19e-05	-0.0122	1.94e-04	0.0817	0.9159
ALL	ComBat-joint	MTR	246.0	0.0426	0.9583	3.51e-04	-0.0118	3.45e-04	6.20e-04	0.7995
ALL	ComBat-joint	MTsat	246.0	0.0619	0.9400	5.09e-04	-0.0116	1.97e-04	0.0040	0.7919
ALL	ComBat-joint	FA	246.0	0.0392	0.9615	3.23e-04	-0.0119	0.0071	0.0102	0.8297
ALL	ComBat-joint	MD	246.0	0.0339	0.9667	2.79e-04	-0.0119	0.0384	0.0160	0.6842
ALL	ComBat-joint	AD	246.0	0.0127	0.9874	1.05e-04	-0.0122	0.0466	0.0159	0.5033
ALL	ComBat-joint	RD	246.0	0.0430	0.9579	3.54e-04	-0.0118	0.0307	0.0137	0.7973

ALL	CovBat	T2w_CSA	246.0	0.0130	0.9871	1.07e-04	-0.0122	5.35e-04	0.1212	0.9348
ALL	CovBat	GM_CSA_mm2	246.0	0.0122	0.9879	1.00e-04	-0.0122	1.42e-04	0.0898	0.8558
ALL	CovBat	MTR	246.0	0.0106	0.9895	8.70e-05	-0.0122	1.23e-04	6.06e-04	0.7606
ALL	CovBat	MTsat	246.0	0.0526	0.9488	4.33e-04	-0.0117	3.80e-04	0.0040	0.7780
ALL	CovBat	FA	246.0	0.0484	0.9527	3.99e-04	-0.0117	0.0057	0.0110	0.8213
ALL	CovBat	MD	246.0	0.0511	0.9502	4.20e-04	-0.0117	0.0400	0.0173	0.6775
ALL	CovBat	AD	246.0	0.0547	0.9468	4.50e-04	-0.0117	0.0528	0.0177	0.4847
ALL	CovBat	RD	246.0	0.0534	0.9480	4.39e-04	-0.0117	0.0307	0.0149	0.7901
ALL	RELIEF	T2w_CSA	246.0	0.0122	0.9879	1.00e-04	-0.0122	4.33e-04	0.1246	0.9361
ALL	RELIEF	GM_CSA_mm2	246.0	0.0016	0.9984	1.34e-05	-0.0123	2.19e-04	0.0921	0.8592
ALL	RELIEF	MTR	246.0	0.0388	0.9620	3.19e-04	-0.0119	3.47e-04	6.23e-04	0.7935
ALL	RELIEF	MTsat	246.0	0.0442	0.9568	3.63e-04	-0.0118	2.02e-04	0.0041	0.7811
ALL	RELIEF	FA	246.0	0.0359	0.9648	2.95e-04	-0.0119	0.0072	0.0103	0.8265
ALL	RELIEF	MD	246.0	0.0265	0.9738	2.18e-04	-0.0120	0.0386	0.0160	0.6859
ALL	RELIEF	AD	246.0	0.0366	0.9640	3.01e-04	-0.0119	0.0480	0.0164	0.4930
ALL	RELIEF	RD	246.0	0.0284	0.9720	2.34e-04	-0.0120	0.0310	0.0138	0.8001
ALL	LME	T2w_CSA	267.0	0.0028	0.9972	2.14e-05	-0.0113	9.86e-07	0.0900	0.9789
ALL	LME	GM_CSA_mm2	267.0	0.0041	0.9959	3.08e-05	-0.0113	2.17e-04	0.0788	0.9118
ALL	LME	MTR	254.0	0.0740	0.9287	5.89e-04	-0.0111	0.0321	0.0015	0.9073
ALL	LME	MTsat	249.0	0.0054	0.9946	4.38e-05	-0.0121	0.0030	0.0087	0.8036
ALL	LME	FA	260.0	0.0217	0.9786	1.69e-04	-0.0114	0.0106	0.0105	0.8433
ALL	LME	MD	260.0	0.0115	0.9886	8.92e-05	-0.0115	0.0161	0.0104	0.7565
ALL	LME	AD	260.0	0.0131	0.9870	1.02e-04	-0.0115	0.0104	0.0100	0.5778
ALL	LME	RD	260.0	0.0316	0.9689	2.46e-04	-0.0113	0.0168	0.0090	0.8507

Note: 'Original' rows show baseline metrics before harmonization. Vendor F and p-value from one-way ANOVA (vendor as factor). eta2 = effect size (vendor variance / total variance). ICC = intraclass correlation coefficient (vendor as random effect). Age and Sex R2 = semi-partial R-squared from multiple regression (age and sex as predictors, vendor as covariate). r(pre,post) = Pearson correlation between pre- and post-harmonization values within each vendor, averaged across vendors.